Last time suggested the notion of:

<u>QR Factorization</u>: A is an $m \times n$ real matrix with linearly independent columns (so $m \geq n$ is forced). Want to write $A = QR$ where Q is $m \times n$ with orthonormal columns, having the same column space as A and R being $n \times n$ upper triangular matrix with positive diagonal entries.

How to find Q? As discussed last time, apply Gram-Schmidt process to columns of A (natural since columns of A are linearly independent). Then normalizing those, create Q via making those the those the columns of Q. Then Q will have orthonormal columns and will have the same column space as A (we know this from how Gram-Schmidt process works) so that is what we need for Q.

How to create R? Say columns of Q are: $u_1, u_2, \ldots, u_n$ and columns of A are: $x_1, x_2, \ldots, x_n$. Then write each x_i as a linear combination of the u_j's.

This is possible since they are basis for same subspace, that is, Col Q = Col A. Thus

$$
\begin{aligned}
x_1 &= r_{11} u_1 \\
x_2 &= r_{12} u_1 + r_{22} u_2 \\
&\vdots \\
x_k &= r_{1k} u_1 + r_{2k} u_2 + \ldots + r_{kk} u_k \\
&\vdots \\
x_n &= r_{1n} u_1 + r_{2n} u_2 + \ldots + r_{nn} u_n
\end{aligned}
$$

Quite important though is not only is each x_i such a linear combination, as shown say x_5 would only need $u_1, u_2, \ldots, u_5$, i.e. would not need say u_7.

That is, to write x_k as such only need to use up to u_k.
Why? Because from Gram-Schmidt we know the span of $\{x_1, x_2, \dots, x_k\}$ is same as $\{u_1, u_2, \dots, u_k\}$

Hence x_k can be written as such.

We can also make each $r_{kk} > 0$ as along the way if one is negative, flip its sign along with sign of u_k.
And we will only potentialy flip that sign at the moment we are fixing that particular x_k, not like we will switch one of them then want to switch it back later.

So once it's "fixed" to be positive, it stays as such, and then we move to the next one.

Can do this since replacing vector with its negative won't affect its span or its length (so that keeps linear independence with length 1). After that, take

$$R = \begin{bmatrix} r_{11} & r_{21} & \cdots & r_{n1} \\ 0 & r_{22} & \cdots & r_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & r_{nn} \end{bmatrix}$$

And then we claim $A = QR$.

Note though that x_k only requiring up to u_k in its linear combination expression in the u_j's is absolutely crucial, otherwise R won't be upper triangular!

But why does $A = QR$? Well the k-th column of A is x_k. The k-th column in QR is Q times the k-th column in R, which has entries $r_{1k}, r_{2k}, \ldots, r_{kk}, 0, 0, \ldots, 0$

Writing that out, since the first column of Q, i.e. u_1 has the entries which multiply r_{1k} and beyond, that becomes

$$x_k = r_{1k}\, u_1 + r_{2k}\, u_2 + \ldots + r_{kk}\, u_k$$

which is indeed the same as it was defined earlier.
Hence why this does work.

Ok that's how to "create" Q and R. But how to "find" them in the most convenient way? Mostly R really.

With Q, as said, use Gram-Schmidt, not really better way usually. But what about R? Solve that system above?

We can, that's not so bad since it's always triangular, not so terrible, but still feels we can do better.

Since $A = QR$, could we left multiply by Q^{-1} maybe to solve for R?

That might be tempting but no! Remember Q is $m \times n$, usually not square so inverse of it won't typically even make sense.

Remember though Q has orthonormal columns so $Q^T Q = I$. Thus left multiply $A = QR$ by Q^T is key. Really accomplishes same thing as you wanted anyway if you were hoping to multiply by Q^{-1}. This gives

$$Q^T A = Q^T QR = IR = R$$

So $R = Q^T A$ is usually most convenient way to find R.

6.4.13

$$A = \begin{bmatrix} 5 & 9 \\ -1 & 7 \\ -3 & -5 \\ 1 & 5 \end{bmatrix}$$

Find QR Factorization of A.

Calling columns of A as x_1, x_2, to make an orthogonal basis choose $v_1 = x_1$, then via Gram-Schmidt:

$$v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1$$

$$= \begin{bmatrix} 9 \\ 7 \\ -5 \\ 5 \end{bmatrix} - \frac{72}{36} \begin{bmatrix} 5 \\ 1 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 5 \\ -3 \\ 1 \end{bmatrix}$$

So using v_1, v_2 (normalizing) we get

$$Q = \frac{1}{6} \begin{bmatrix} 5 & -1 \\ -1 & 5 \\ -3 & -3 \\ 1 & 1 \end{bmatrix}$$

So then

$$R = \frac{1}{6} \begin{bmatrix} 5 & 1 & -3 & 1 \\ -1 & 5 & 1 & 3 \end{bmatrix} \begin{bmatrix} 5 & 9 \\ -1 & 7 \\ -3 & -5 \\ 1 & 5 \end{bmatrix}$$

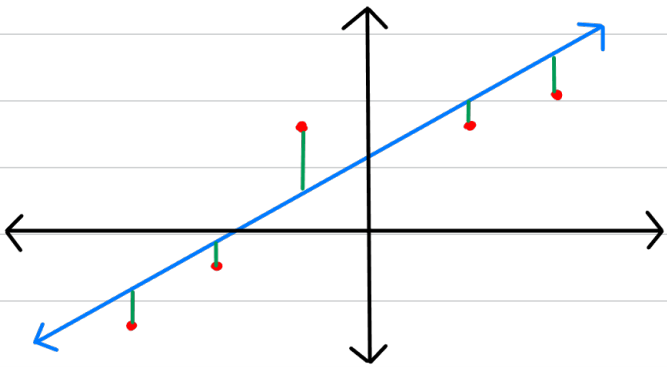
----> $$R = \frac{1}{6}\begin{bmatrix} 36 & 72 \\ 0 & 36 \end{bmatrix} = \begin{bmatrix} 6 & 12 \\ 0 & 6 \end{bmatrix}$$

Notice R is upper triangular with positive diagonal entries as it's supposed to be.

Note that in this question they actually gave you Q, didn't expect you to compute it as I did. In general you should expect to have to compute Q though.

Here it's not so bad since Q only has two vectors so the Gram-Schmidt process isn't too ugly. With three or more it becomes much worse, especially since even with three vectors you will presumably be in minimum $\mathbb{R}^4$ (so that the vectors don't generate the full space). And that pretty much will never be trivial.

6.5 Least Squares



Blue line is least squares line. That is, those five red points, amongst all lines, for which is the sum of the squares of those vertical distances minimized?

Why minimize the sum of squares of distances rather rather than sums of distances themselves? Two reasons

(1) Using actual distance, when modeling you have to worry about whether point is higher or line is higher when subtracting. So since might be fairly mixed, way to deal with that would be to use absolute values. But absolute values are much more annoying to deal

with in Calculus (say with derivatives) compared to squares. Squaring avoids the issue of having to worry about whether point or line is higher.

(2) Bigger reason is if we want to expand on this, say in higher dimensions, distance formula uses squares, distance being absolute value really only works when the distance in question is one dimensional.

You may have seen a formula for this in a prior Calculus or Statistics class. But likely only the formula, not the justification for it.

What if we decided a line won't be accurate enough, we say want the "best fitting parabola" i.e. "least squares parabola" since we want a better fitting curve. Naturally degree two gives us more flexibility than degree one so that makes sense.

Of course that will be harder to compute as well, the usual dilemma that we can try to make it more accurate, at the cost of doing more calculation.

Whether such a question will be reasonable to do by hand, important to understand how we would obtain it if we wanted to say teach a machine to find it.

So we try to justify how least squares is obtained, whether line or parabola. But ok, actually doing that is more the next section. This section more sets the stage for how that relates to linear algebra ideas that we studied with projections.

It makes sense that such ideas could be involved as those involved ideas of "closest point" as well as "best approximation" which certainly sounds similar to what least squares is attempting to accomplish.

Least squares line is natural thing to use when you have many data points. Likely not a simple function which will contain all those points so what to do?

If we want to get a reasonable idea of how that data is working without doing tons of calculation, least squares line might be good way since line is the simplest type of function there is.

Say A is $m \times n$ real matrix and $b \in \mathbb{R}^m$. As usual, think of A as being a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ Note that domain and codomain of A are usually not the same.

We are quite familiar with concept of solving matrix equation $Ax = b$. But if that were inconsistent, i.e. b was not in the range of A, is there a point in the range of A which is closest to b?

E.g. say we had a set of points, and we wanted to find a line passing through all of them. Might set up a system to try to solve for the line, but if there are more than two points, might not be a line passing through them all, thus a situation where you might wind up with such an inconsistent system.

Since range of A is a subspace of $\mathbb{R}^m$, namely Col A, yes! It will be the projection of b on Col A, which we will as usual denote as $\hat{b}$. So that means overall want to find all $\hat{x} \in \mathbb{R}^n$ such that

$$A\hat{x} = \hat{b}$$

or in other words all $\hat{x}$ for which $\| b - A\hat{x} \|$ is as small as possible.

Important to note that $\hat{b}$ is unique, we proved that about projections earlier. However there could be multiple $\hat{x}$ which map to $\hat{b}$. Whether there are or not is something we will want to investigate in this.

Since $b \in \text{Col } A$, there will be at least one which works, unlike $Ax = b$ which was probably inconsistent.

Probably because if it was consistent, then the projection would just be b itself. Fine but not very interesting.

So then question is how to find all value(s) of $\hat{x}$ for which $A\hat{x} = \hat{b}$, including whether $\hat{x}$ is unique or not.

In theory if we brute forced this, the steps would be:

(1) Find a basis for Col A via putting A in REF and using the pivot columns.
(2) Turn that basis orthogonal via Gram-Schmidt.
(3) Use projection formula to find $\hat{b}$
(4) Solve system $A\hat{x} = \hat{b}$ via row reduction.

That seems very long though! Can we simplify it? Indeed we can. Not that this makes it easy but usually much better than the above. Big key is

$$b - \hat{b} = b - A\hat{x}$$

must be orthogonal to Col A by earlier ideas with projections. Thus we must have

$$A^T(b - A\hat{x}) = 0$$

for this to work.

This is because orthogonal to Col A is equivalent to orthogonal to Row (A^T) which means being in Nul (A^T) and that is exactly what that equation reads as. Hence rearranging that gives

$$A^T A\hat{x} = A^T b$$

so key is to find all $\hat{x}$ satisfying this. Thus if given A, b

compute $A^T A$, $A^T b$ then solve system for $\hat{x}$.

6.5.1

$$A = \begin{bmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

Find the point in Col A which is closest to b.

Use the calculation ideas mentioned earlier:

$$A^T A = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -3 & 3 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 6 & -11 \\ -11 & 22 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -3 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ 11 \end{bmatrix}$$

We need to solve

$$A^T A \hat{x} = A^T b$$

which from the above is equivalent to:

$$\begin{bmatrix} 6 & -11 \\ -11 & 22 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -4 \\ 11 \end{bmatrix}$$

and that becomes

$$6x_1 - 11x_2 = -4$$
$$-x_1 + 2x_2 = 1$$

$$x_2 = 2 \text{ ----> } x_1 = 3$$

Thus $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ is the unique solution here.

Hence

$$A\hat{x} = \begin{bmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$$

is the point in Col A which is closest to b.

Ok this wasn't so bad, but as usual if matrices get a bit bigger, can get much more extreme quickly.

And, as mentioned, when is solution unique versus not?

Again goal is to solve

$$A^T A\hat{x} = A^T b$$

Can we make this easier? Maybe left multiply by $(A^T)^{-1}$ to cancel term in front? Unfortunately can't do that since may not be square matrix so inverse won't even make sense most of the time!

Same type of issue we ran into with $A = QR$ when trying to solve for R. What about left multiplying by $(A^T A)^{-1}$? Well $A^T A$ is always square so that is potentially more viable. But is it necessarily invertible?

That becomes an important issue in all this. So need to investigate that. Turns out QR Factorization is also potentially a help in making this easier to solve.